



HNDIT4242-Computer Services Management



Lecture 10 Capacity Planning

Lesson Objectives

- Understand the **importance of capacity planning** in IT infrastructure.
- Learn how to **monitor and analyze IT resource usage** for performance optimization.
- Develop a **scalable IT investment strategy** for future growth.
- Implement **security policies** to ensure IT systems remain protected as they scale.
- Explore **best practices for IT capacity management** to prevent system overload.



1. Understanding IT Capacity Planning

1.1 What is IT Capacity Planning?

- **Definition:** The process of ensuring that IT infrastructure meets current and future business needs.
- **Why It's Important:**
 - ✓ Prevents **performance bottlenecks**.
 - ✓ Ensures **cost efficiency** by optimizing resource allocation.
 - ✓ Supports **business growth** by scaling IT resources effectively.
 - ✓ Enhances **system reliability** and uptime.



1.2 Key Components of Capacity Planning

- **Workload Forecasting** – Predicting future system demands.
- **Performance Analysis** – Evaluating existing IT resource utilization.
- **Scalability Planning** – Designing IT systems to handle growth.
- **Cost Management** – Budgeting IT investments wisely.
- **Security & Compliance** – Ensuring secure growth of IT infrastructure.



2. Monitoring IT Resource Usage

2.1 IT Resource Monitoring Metrics

- **CPU Usage** – Measures processor workload.
- **Memory Utilization** – Tracks RAM consumption.
- **Storage Usage** – Analyzes disk space availability.
- **Network Bandwidth** – Monitors data transfer rates.
- **Response Time** – Evaluates system efficiency in handling requests.

2.2 Tools for IT Resource Monitoring

Tool	Purpose
Nagios	Server & network monitoring
Zabbix	Real-time performance tracking
SolarWinds	Advanced IT infrastructure monitoring
AWS CloudWatch	Cloud resource monitoring
New Relic	Application performance management



3. Developing a Technology Strategy & Investment Plan

3.1 IT Investment Planning

- **Why It Matters:** Proper budgeting helps avoid overspending while ensuring systems stay up to date.
- **Steps in IT Investment Planning:**
 - **Assess current IT infrastructure** – Identify gaps and inefficiencies.
 - **Estimate future IT needs** – Based on growth projections.
 - **Prioritize investments** – Focus on critical upgrades first.
 - **Plan for scalability** – Invest in flexible and scalable solutions.
 - **Ensure ROI (Return on Investment)** – Choose solutions that bring long-term value.



3.2 Scalability Considerations in IT Strategy

- **Horizontal Scaling (Scaling Out)** – Adding more servers to distribute workloads.
- **Vertical Scaling (Scaling Up)** – Increasing CPU, RAM, or storage on existing servers.
- **Cloud Computing Solutions** – Using **AWS, Azure, or Google Cloud** for on-demand scaling.



3.3 Cost Estimation for IT Projects

Cost Factor	Example
Hardware	Servers, storage devices, networking equipment
Software	Licensing fees for operating systems, databases, applications
Cloud Services	AWS, Azure, Google Cloud pricing models
Maintenance	IT support, system upgrades, and backups
Security	Firewalls, antivirus, compliance audits



4. Security Considerations in Capacity Planning

4.1 Importance of Security in IT Expansion

- As IT systems scale, **cybersecurity risks increase.**
- Capacity planning must ensure **new resources are secure and comply with industry standards.**



4.2 Security Policies for Scalable IT Systems

- **Access Control & Authentication** – Implement role-based access (**RBAC**) and multi-factor authentication (**MFA**).
- **Data Encryption** – Use end-to-end encryption for sensitive data.
- **Regular Security Audits** – Assess system vulnerabilities.
- **Incident Response Plans** – Prepare for **cyber threats** and system failures.
- **Compliance with Standards** – Follow **GDPR, ISO 27001, NIST** security frameworks.



4.3 Cloud Security in Capacity Planning

- Cloud scalability requires strong security measures:
 - ✓ **IAM (Identity & Access Management)** – Restricting permissions.
 - ✓ **DDoS Protection** – Preventing service disruptions.
 - ✓ **Backup & Disaster Recovery** – Ensuring business continuity.



Summary

- **Capacity planning** ensures IT systems are scalable and perform efficiently.
- **Monitoring IT resource usage** helps prevent system failures and optimize performance.
- **Strategic IT investment** ensures cost-effective expansion while maintaining service quality.
- **Security policies** are critical when scaling IT infrastructure to protect data and compliance.



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